Chapter 1

Vector Analysis

Problem 1.1

Show that the dot product and cross product is distributive

Solution:

Let the three vectors be

$$\boldsymbol{v}_1 = [a, b, c]$$
$$\boldsymbol{v}_2 = [d, e, f]$$
$$\boldsymbol{v}_3 = [g, h, i]$$

dot product:

Show

$$\boldsymbol{v}_1 \cdot (\boldsymbol{v}_2 + \boldsymbol{v}_3) = \boldsymbol{v_1} \cdot \boldsymbol{v}_2 + \boldsymbol{v_1} \cdot \boldsymbol{v}_3$$

Using

$$\boldsymbol{v}_2 + \boldsymbol{v}_3 = [d + g, e + h, f + i]$$
$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = ad + be + cf$$
$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_3 = ag + bh + ci$$

LHS:

$$\boldsymbol{v}_1 \cdot (\boldsymbol{v}_2 + \boldsymbol{v}_3) = [a, b, c] \cdot [d + g, e + h, f + i] =$$
$$= a(d + g) + b(e + h) + c(f + i)$$

RHS:

$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 + \boldsymbol{v}_1 + \boldsymbol{v}_3 = ad + be + cf + ag + bh + ci =$$
$$= a(d + g) + b(e + h) + c(f + i)$$

cross product: Show

$$\boldsymbol{v}_1 \times (\boldsymbol{v}_2 + \boldsymbol{v}_3) = (\boldsymbol{v}_1 \times \boldsymbol{v}_2) + (\boldsymbol{v}_1 \times \boldsymbol{v}_3)$$

Using

$$\boldsymbol{v}_2 + \boldsymbol{v}_3 = [d+g, e+h, f+i]$$
$$\boldsymbol{v}_1 \times \boldsymbol{v}_2 = bf - ce + cd - af + ae - bd$$
$$\boldsymbol{v}_1 \times \boldsymbol{v}_3 = bi - ch + cg - ai + ah - bg$$

which gives

$$(\boldsymbol{v}_1 \times \boldsymbol{v}_2) + (\boldsymbol{v}_1 \times \boldsymbol{v}_3) = a(e+h-f-i) + b(f+i-d-g) + c(d+g-e-h)$$

and LHS:

$$\begin{aligned}\boldsymbol{v}_1 \times (\boldsymbol{v}_2 + \boldsymbol{v}_3) &= b(f+i) - c(e+h) + c(d+g) - a(f+i) + a(e+h) - b(d+g) = \\ &= bf + bi - ce - ch + cd + cg - af - ai + ae + ah - bd - bg = \\ &= a(e+h-f-i) + b(f+i-d-g) + c(d+g-e-h)\end{aligned}$$

□

Problem 1.2

Is the cross product associative?

$$(\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C} \stackrel{?}{=} \boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C})$$

Solution:

No, for example set

$$\boldsymbol{A} = \hat{\boldsymbol{x}}$$
$$\boldsymbol{B} = \hat{\boldsymbol{x}}$$
$$\boldsymbol{C} = \hat{\boldsymbol{z}}$$

Then, the left hand side above becomes

$$(\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C} = (\hat{\boldsymbol{x}} \times \hat{\boldsymbol{x}}) \times \hat{\boldsymbol{z}} = 0$$

while the right hand side is

$$\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \hat{\boldsymbol{x}} \times (\hat{\boldsymbol{x}} \times \hat{\boldsymbol{z}}) = \hat{\boldsymbol{x}} \times -\hat{\boldsymbol{y}} = -\hat{\boldsymbol{z}}$$

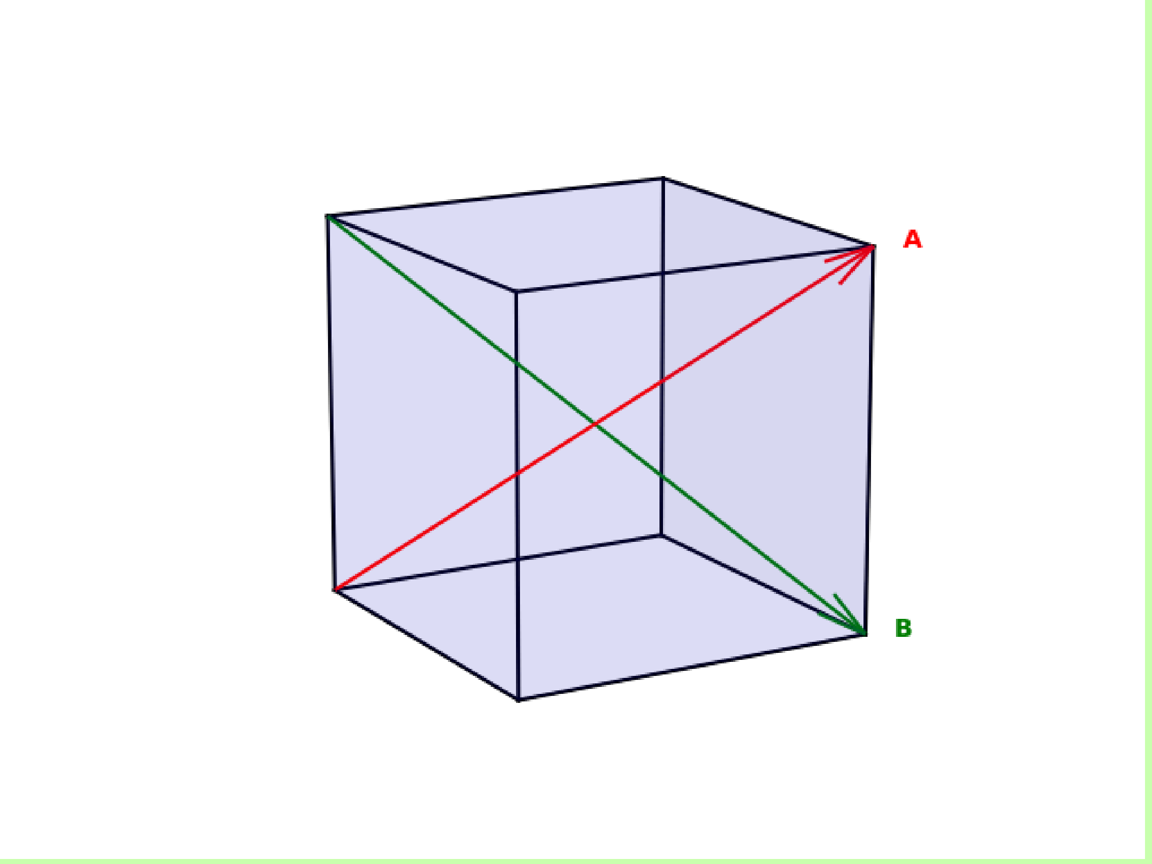
□

Problem 1.3

Find the angle between the body diagonals of a cube.

Solution:

The body diagonals $\boldsymbol{A} = [1,1,1]$ and $\boldsymbol{B} = [1,1,-1]$



$|\boldsymbol{A}| = |\boldsymbol{B}| = \sqrt{3}$

$\boldsymbol{A} \cdot \boldsymbol{B} = 1 + 1 - 1 = 1$

$\boldsymbol{A} \cdot \boldsymbol{B} = |\boldsymbol{A}|\ |\boldsymbol{B}| \cos\theta = 3\cos\theta = 1 \to \theta = \arccos\left(\frac{1}{3}\right) \approx 70.53° \ \square$

Problem 1.4

Use the cross product to find the components of the unit vector $\hat{\boldsymbol{n}}$ perpendicular to the shaded plane in Fig.

Solution:

Two of the plane vectors are $\boldsymbol{v}_1 = [-1,2,0]$ and $\boldsymbol{v}_2 = [-1,0,3]$. $\boldsymbol{v}_1 \times \boldsymbol{v}_2 = n = [6,3,2]$ and $|\boldsymbol{n}| = 7$ gives $\hat{\boldsymbol{n}} = \frac{1}{7}[6,3,2]$ $\square$

Problem 1.5

Prove the **BAC-CAB** rule,

$$\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B})$$

by writing out both sides in component form.

Solution:

Writing out the vectors in component form

$$\begin{aligned} \boldsymbol{A} &= A_x \hat{\boldsymbol{x}} + A_y \hat{\boldsymbol{y}} + A_z \hat{\boldsymbol{z}} \\ \boldsymbol{B} &= B_x \hat{\boldsymbol{x}} + B_y \hat{\boldsymbol{y}} + B_z \hat{\boldsymbol{z}} \\ \boldsymbol{C} &= C_x \hat{\boldsymbol{x}} + C_y \hat{\boldsymbol{y}} + C_z \hat{\boldsymbol{z}} \end{aligned}$$

starting with lhs

$$\boldsymbol{B} \times \boldsymbol{C} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} = \left(B_y C_z - B_z C_y\right)\hat{\boldsymbol{x}} + (B_z C_x - B_x C_z)\hat{\boldsymbol{y}} + \left(B_x C_y - B_y C_x\right)\hat{\boldsymbol{z}}$$

rewriting with a new matrix $\boldsymbol{D}$ where

$$\begin{aligned} D_x &= B_y C_z - B_z C_y \\ D_y &= B_z C_x - B_x C_z \\ D_z &= B_x C_y - B_y C_x \end{aligned}$$

gives

$$\boldsymbol{D} = \boldsymbol{B} \times \boldsymbol{C} = D_x \hat{\boldsymbol{x}} + D_y \hat{\boldsymbol{y}} + D_z \hat{\boldsymbol{z}}$$

Now

$$\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{A} \times \boldsymbol{D} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ A_x & A_y & A_z \\ D_x & D_y & D_z \end{vmatrix} =$$

$$= \left(A_y D_z - A_z D_y\right)\hat{\boldsymbol{x}} + (A_z D_x - A_x D_z)\hat{\boldsymbol{y}} + \left(A_x D_y - A_y D_x\right)\hat{\boldsymbol{z}}$$

RHS:

$$\begin{aligned} \boldsymbol{A} \cdot \boldsymbol{C} &= A_x C_x + A_y C_y + A_z C_z = P \\ \boldsymbol{A} \cdot \boldsymbol{B} &= A_x B_x + A_y B_y + A_z B_z = Q \\ \boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) &= B_x P \hat{\boldsymbol{x}} + B_y P \hat{\boldsymbol{y}} + B_z P \hat{\boldsymbol{z}} \\ \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B}) &= C_x Q \hat{\boldsymbol{x}} + C_y Q \hat{\boldsymbol{y}} + C_z Q \hat{\boldsymbol{z}} \\ \boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B}) &= (B_x P - C_x Q)\hat{\boldsymbol{x}} + \left(B_y P - C_y Q\right)\hat{\boldsymbol{y}} + (B_z P - C_z Q)\hat{\boldsymbol{z}} \end{aligned}$$

Now final step is checking if

$$\mathbf{1}\text{: } A_y D_z - A_z D_y = B_x P - C_x Q$$
$$\mathbf{2}\text{: } A_z D_x - A_x D_z = B_y P - C_y Q$$
$$\mathbf{3}\text{: } A_x D_y - A_y D_x = B_z P - C_z Q$$

For **1**:

$$\text{LHS: } A_y\left(B_x C_y - B_y C_x\right) - A_z\left(B_z C_x - B_x C_z\right)$$
$$= A_y B_x C_y - A_y B_y C_x - A_z B_z C_x + A_z B_x C_z$$
$$= B_x\left(A_y C_y + A_z C_z\right) - C_x\left(A_y B_y + A_z B_z\right)$$

$$\text{RHS: } B_x\left(A_x C_x + A_y C_y + A_z C_z\right) - C_x\left(A_x B_x + A_y B_y + A_z B_z\right)$$
$$= B_x A_x C_x + B_x\left(A_y C_y + A_z C_z\right) - C_x A_x B_x - C_x\left(A_y B_y + A_z B_z\right)$$
$$= B_x\left(A_y C_y + A_z C_z\right) - C_x\left(A_y B_y + A_z B_z\right) = \text{LHS} \checkmark$$

For **2**:

$$\text{LHS: } A_z\left(B_y C_z - B_z C_y\right) - A_x\left(B_x C_y - B_y C_x\right)$$
$$= A_z B_y C_z - A_z B_z C_y - A_x B_x C_y + A_x B_y C_x$$
$$= B_y(A_z C_z + A_x C_x) - C_y(A_z B_z + A_x B_x)$$

$$\text{RHS: } B_y\left(A_x C_x + A_y C_y + A_z C_z\right) - C_y\left(A_x B_x + A_y B_y + A_z B_z\right)$$
$$= B_y A_y C_y + B_y(A_x C_x + A_z C_z) - C_y A_y B_y - C_y(A_x B_x + A_z B_z)$$
$$= B_y(A_x C_x + A_z C_z) - C_y(A_x B_x + A_z B_z) = \text{LHS} \checkmark$$

For **3**:

$$\text{LHS: } A_x(B_z C_x - B_x C_z) - A_y\left(B_y C_z - B_z C_y\right)$$
$$= A_x B_z C_x - A_x B_x C_z - A_y B_y C_z + A_y B_z C_y$$
$$= B_z\left(A_x C_x + A_y C_y\right) - C_z\left(A_x B_x + A_y B_y\right)$$

$$\text{RHS: } B_z\left(A_x C_x + A_y C_y + A_z C_z\right) - C_z\left(A_x B_x + A_y B_y + A_z B_z\right)$$
$$= B_z A_z C_z + B_z\left(A_x C_x + A_y C_y\right) - C_z A_z B_z - C_z\left(A_x B_x + A_y B_y\right)$$
$$= B_z\left(A_x C_x + A_y C_y\right) - C_z\left(A_x B_x + A_y B_y\right) = \text{LHS} \checkmark$$

Since all three components are equal, the BAC-CAB rule is proven. □

Problem 1.6

Prove that

$$[\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C})] + [\boldsymbol{B} \times (\boldsymbol{C} \times \boldsymbol{A})] + [\boldsymbol{C} \times (\boldsymbol{A} \times \boldsymbol{B})] = 0$$

Under what conditions does $\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = (\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C}$?

Solution:

Using the BAC-CAB rule, rewrite the triple cross product as

$$
\begin{aligned}
&(1)\ \boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B}) \\
&(2)\ \boldsymbol{B} \times (\boldsymbol{C} \times \boldsymbol{A}) = \boldsymbol{C}(\boldsymbol{B} \cdot \boldsymbol{A}) - \boldsymbol{A}(\boldsymbol{B} \cdot \boldsymbol{C}) \\
&(3)\ \boldsymbol{C} \times (\boldsymbol{A} \times \boldsymbol{B}) = \boldsymbol{A}(\boldsymbol{C} \cdot \boldsymbol{B}) - \boldsymbol{B}(\boldsymbol{C} \cdot \boldsymbol{A})
\end{aligned}
$$

Using the commutative property of the dot product, $\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = \boldsymbol{v}_2 \cdot \boldsymbol{v}_1$, it is seen that:

- The $\boldsymbol{B}$-part in (1) cancels out the $\boldsymbol{B}$-part in (3)
- The $\boldsymbol{C}$-part in (1) cancels out the $\boldsymbol{C}$-part in (2)
- The $\boldsymbol{A}$-part in (2) cancels out the $\boldsymbol{A}$-part in (3)

The conditions when $\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = (\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C}$ is translated with the BAC-CAB rule into $\boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B}) = -\boldsymbol{A}(\boldsymbol{C} \cdot \boldsymbol{B}) + \boldsymbol{B}(\boldsymbol{C} \cdot \boldsymbol{A})$ which gives that either $\boldsymbol{A} = \boldsymbol{C}$ or $\boldsymbol{B} \perp (\boldsymbol{A} \text{ and } \boldsymbol{C})$ $\square$

Problem 1.7

Find the separation vector $\mathscr{r}\, rs$ from the source point (2, 8, 7) to the field point (4, 6, 8). Determine its magnitude ($\mathscr{r}$), and construct the unit vector $\hat{\boldsymbol{\mathscr{r}}}$.

Solution:

The separation vector $\boldsymbol{\mathscr{r}}$ is the difference between the field point and the source point.

$$\boldsymbol{\mathscr{r}} = (4, 6, 8) - (2, 8, 7) = (2, -2, 1)$$

The magnitude is

$$|\boldsymbol{\mathscr{r}}| = \mathscr{r} = \sqrt{2^2 + (-2)^2 + 1^2} = 3$$

and the unit vector is

$$\frac{\boldsymbol{\mathscr{r}}}{\mathscr{r}} = \hat{\boldsymbol{\mathscr{r}}} = \frac{1}{3}(2, -2, 1)$$

$\square$

Problem 1.8

Problem 1.9

Problem 1.10

Problem 1.11

Find the gradients of the following functions:

(a) $f(x, y, z) = x^2 + y^3 + z^4$

(b) $f(x, y, z) = x^2 y^3 z^4$

(c) $f(x, y, z) = e^x \sin(y) \ln(z)$

Solution:

The gradient of a function T is defined as

$$\boldsymbol{\nabla} T = \frac{\partial T}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial T}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial T}{\partial z}\hat{\boldsymbol{z}}$$

This results in

(a) $2x\hat{\boldsymbol{x}} + 3y^2\hat{\boldsymbol{y}} + 4z^3\hat{\boldsymbol{z}}$

(b) $2xy^3z^4\hat{\boldsymbol{x}} + 3x^2y^2z^4\hat{\boldsymbol{y}} + 4x^2y^3z^3\hat{\boldsymbol{z}}$

(c) $e^x\left(\sin(y)\ln(z)\hat{\boldsymbol{x}} + \cos(y)\ln(z)\hat{\boldsymbol{y}} + \frac{\sin(y)}{z}\hat{\boldsymbol{z}}\right)$

□

Problem 1.12

The height of a certain hill (in feet) is given by

$$h(x, y) = 10(2xy - 3x^2 - 4y^2 - 18x + 28y + 12)$$

where y is the distance (in miles) north, x the distance east, of South Hadley.

(a) Where is the top of the hill located?

(b) How high is the hill?

(c) How steep is the slope (in feet per mile) at a point 1 mile north and 1 mile east of South Hadley? In what direction is the slope steepest, at that point?

Solution:

The gradient of $h(x, y)$ is

$$\nabla h = (20y - 60x - 180)\hat{\boldsymbol{x}} + (20x - 80y + 280)\hat{\boldsymbol{y}}$$

(a) The location of the top is calculated by setting the gradient to 0. This means

$$20y - 60x - 180 = 0$$
$$20x - 80y + 280 = 0$$

which has the solution $(x, y) = (-2, 3)$ miles.

(b) Putting the values given by (a) into the height function $h(x, y)$, the height of the hill is $h(-2, 3) = 720\text{ft}$.

(c) At the point $(x, y) = (1, 1)$, the steepness and direction is obtained by putting this point into the gradient. The slope is then $\sqrt{2} \cdot 220$ ft/mile in the direction $-\hat{\boldsymbol{x}} + \hat{\boldsymbol{y}}$, which corresponds to northwest. □

Problem 1.13

Let $\boldsymbol{\mathscr{r}}$ be the separation vector from a fixed point (x', y', z') to the point (x, y, z), and let $\mathscr{r}$ be its length. Show that

(a) $\nabla(\mathscr{r}^2) = 2\,\boldsymbol{\mathscr{r}}$

(b) $\nabla\left(\frac{1}{\mathscr{r}}\right) = -\hat{\boldsymbol{\mathscr{r}}}/\mathscr{r}^2$

(c) What is the general formula for $\nabla(\mathscr{r}^n)$?

Solution:

$$\boldsymbol{\mathscr{r}} = (x - x', y - y', z - z')$$
$$\mathscr{r} = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}$$

(a)

$$\nabla(\mathscr{r}^2) = \nabla\Big((x - x')^2 + (y - y')^2 + (z - z')^2\Big) =$$
$$= 2(x - x')\hat{\boldsymbol{x}} + 2(y - y')\hat{\boldsymbol{y}} + 2(z - z')\hat{\boldsymbol{z}} = 2\,\boldsymbol{\mathscr{r}}$$

(b)

$$\nabla\left(\frac{1}{\mathscr{r}}\right) = \nabla\left(\frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}\right) =$$

$$= -\frac{1}{2}\frac{2(x-x')+2(y-y')+2(z-z')}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}^3} = -\frac{\boldsymbol{\mathscr{r}}}{\mathscr{r}^3} = -\frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}$$

(c)

$$\nabla(\mathscr{r}^n) = \nabla\left(\left((x-x')^2+(y-y')^2+(z-z')^2\right)^{\frac{n}{2}}\right) =$$

$$= \frac{n}{2}(2(x-x')\hat{\boldsymbol{x}}+2(y-y')\hat{\boldsymbol{y}}+2(z-z')\hat{\boldsymbol{z}})\left[(x-x')^2+(y-y')^2+(z-z')^2\right]^{\frac{n-2}{2}} =$$

$$= n\mathscr{r}^{n-2}\ \boldsymbol{\mathscr{r}}$$

□

Problem 1.14

Problem 1.15

Calculate the divergence of the following vector functions:

(a) $\boldsymbol{v}_a = x^2\hat{\boldsymbol{x}} + 3xz^2\hat{\boldsymbol{y}} - 2xz\hat{\boldsymbol{z}}$

(b) $\boldsymbol{v}_b = xy\hat{\boldsymbol{x}} + 2yz\hat{\boldsymbol{y}} + 3xz\hat{\boldsymbol{z}}$

(c) $\boldsymbol{v}_c = y^2\hat{\boldsymbol{x}} + (2xy+z^2)\hat{\boldsymbol{y}} + 2yz\hat{\boldsymbol{z}}$

Solution:

The divergence of a vector $\boldsymbol{v}$ is defined as

$$\nabla\cdot\boldsymbol{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

$$\text{(a) } \nabla\cdot\boldsymbol{v}_a = 2x - 2x = 0$$
$$\text{(b) } \nabla\cdot\boldsymbol{v}_b = y + 2z + 3x$$
$$\text{(c) } \nabla\cdot\boldsymbol{v}_c = 2x + 2y$$

□

Problem 1.16

Sketch the vector function

$$\boldsymbol{v} = \frac{\hat{\boldsymbol{r}}}{r^2},$$

and compute its divergence.

Solution:

The divergence of the r-component in spherical coordinates is

$$\boldsymbol{\nabla} \cdot \boldsymbol{v} = \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 v_r)$$

Using this, the divergence is $\boldsymbol{\nabla} \cdot \boldsymbol{v} = 0$. From the sketch, clearly the divergence is not 0 at the origin. The problem arises at the point $\boldsymbol{r} = 0$ where $\boldsymbol{v} \to \infty$. At all other points $\boldsymbol{r} \neq 0$, the divergence is 0.

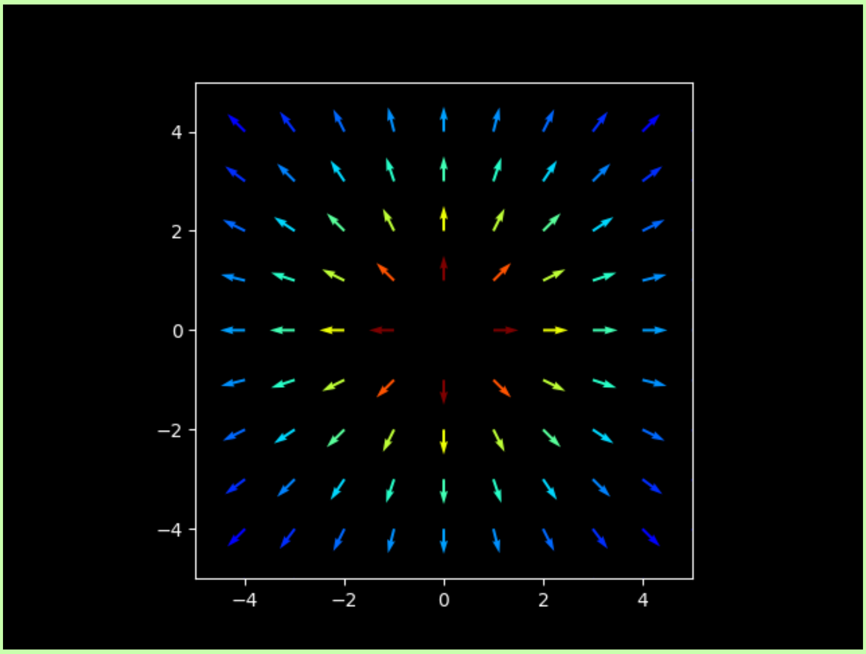


Figure 2: Sketch of the function $\boldsymbol{v} = \frac{\hat{\boldsymbol{r}}}{r^2}$

□

Problem 1.17

Problem 1.18

Calculate the curls of the vector functions

(a) $\boldsymbol{v}_a = x^2\hat{\boldsymbol{x}} + 3xz^2\hat{\boldsymbol{y}} - 2xz\hat{\boldsymbol{z}}$

(b) $\boldsymbol{v}_b = xy\hat{\boldsymbol{x}} + 2yz\hat{\boldsymbol{y}} + 3xz\hat{\boldsymbol{z}}$

(c) $\boldsymbol{v}_c = y^2\hat{\boldsymbol{x}} + (2xy + z^2)\hat{\boldsymbol{y}} + 2yz\hat{\boldsymbol{z}}$

Solution:

The definition of the curl of a vector function $\boldsymbol{v}$ is

$$\boldsymbol{\nabla} \times \boldsymbol{v} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\boldsymbol{z}}$$

This gives

(a) $\boldsymbol{\nabla} \times \boldsymbol{v}_a = -6xz\hat{\boldsymbol{x}} + 2z\hat{\boldsymbol{y}} + 3z^2\hat{\boldsymbol{z}}$

(b) $\boldsymbol{\nabla} \times \boldsymbol{v}_b = -2y\hat{\boldsymbol{x}} + -3z\hat{\boldsymbol{y}} + -x\hat{\boldsymbol{z}}$

(c) $\boldsymbol{\nabla} \times \boldsymbol{v}_c = 0$

$\square$

Problem 1.19

Problem 1.20

Construct a vector function that has a zero divergence and zero curl everywhere.

Solution:

The vector function $\boldsymbol{R}$ should satisfy

$$\boldsymbol{\nabla} \cdot \boldsymbol{R} = 0$$
$$\boldsymbol{\nabla} \times \boldsymbol{R} = 0$$

Using the definitions of the divergence and curl, the conditions are

$$\boldsymbol{\nabla} \cdot \boldsymbol{R} = \frac{\partial R_x}{\partial x} + \frac{\partial R_y}{\partial y} + \frac{\partial R_z}{\partial z} = 0$$

$$\boldsymbol{\nabla} \times \boldsymbol{R} = \left(\frac{\partial R_z}{\partial y} - \frac{\partial R_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial R_x}{\partial z} - \frac{\partial R_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial R_y}{\partial x} - \frac{\partial R_x}{\partial y}\right)\hat{\boldsymbol{z}}$$

Want to fulfill:

$$\frac{\partial R_z}{\partial y} = \frac{\partial R_y}{\partial z}$$
$$\frac{\partial R_x}{\partial z} = \frac{\partial R_z}{\partial x}$$
$$\frac{\partial R_y}{\partial x} = \frac{\partial R_x}{\partial y}$$

An example is $\boldsymbol{R} = yz\hat{\boldsymbol{x}} + xz\hat{\boldsymbol{y}} + xy\hat{\boldsymbol{z}}$

□

Problem 1.21

Prove the product rules

$$\begin{aligned} &\text{(i)} && \boldsymbol{\nabla}(fg) = f\boldsymbol{\nabla}g + g\boldsymbol{\nabla}f \\ &\text{(iv)} && \boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A}) - \boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B}) \\ &\text{(v)} && \boldsymbol{\nabla}\times(f\boldsymbol{A}) = f(\boldsymbol{\nabla}\times\boldsymbol{A}) - \boldsymbol{A}\times(\boldsymbol{\nabla}f) \end{aligned}$$

Problem 1.22

Problem 1.23

Problem 1.24

Problem 1.25

$$\begin{aligned} &\text{(ii)} && \boldsymbol{\nabla}(\boldsymbol{A}\cdot\boldsymbol{B}) = \boldsymbol{A}\times(\boldsymbol{\nabla}\times\boldsymbol{B}) + \boldsymbol{B}\times(\boldsymbol{\nabla}\times\boldsymbol{A}) + (\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B} + (\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A} \\ &\text{(iv)} && \boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A}) - \boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B}) \\ &\text{(vi)} && \boldsymbol{\nabla}\times(\boldsymbol{A}\times\boldsymbol{B}) = (\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A} - (\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B} + \boldsymbol{A}(\boldsymbol{\nabla}\cdot\boldsymbol{B}) - \boldsymbol{B}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) \end{aligned}$$

(a) Check product rule (iv) for the functions

$$\boldsymbol{A} = x\hat{\boldsymbol{x}} + 2y\hat{\boldsymbol{y}} + 3z\hat{\boldsymbol{z}}\ ,\quad \boldsymbol{B} = 3y\hat{\boldsymbol{x}} - 2x\hat{\boldsymbol{y}}$$

(b) Do the same for product rule (ii).

(c) Do the same for rule (vi).

Solution:

(a) Start with LHS:

$$\boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \boldsymbol{\nabla}\cdot\big((A_yB_z - A_zB_y)\hat{\boldsymbol{x}} + (A_zB_x - A_xB_z)\hat{\boldsymbol{y}} + (A_xB_y - A_yB_x)\hat{\boldsymbol{z}}\big) =$$

$$= \frac{\partial}{\partial x}(A_yB_z - A_zB_y) + \frac{\partial}{\partial y}(A_zB_x - A_xB_z) + \frac{\partial}{\partial z}(A_xB_y - A_yB_x) =$$

$$= \frac{\partial}{\partial x}(6xz) + \frac{\partial}{\partial y}(9zy) + \frac{\partial}{\partial z}(-2x^2 - 6y^2) = 6z + 9z = \boxed{15z}$$

RHS:

$$\underbrace{\boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A})}_{(1)} - \underbrace{\boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B})}_{(2)} =$$

$$(1)\ \boldsymbol{B}\cdot\left[\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{\boldsymbol{z}}\right] =$$

$$= B_x\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + B_y\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + B_z\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) = 0$$

$$(2)\ \boldsymbol{A}\cdot\left[\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right)\hat{\boldsymbol{z}}\right] =$$

$$= A_x\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right) + A_y\left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right) + A_z\left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right) =$$

$$= 3z(-5) = -15z$$

$$(1) - (2) = 0 - (-15z) = \boxed{15z}$$

(b) LHS:

$$\boldsymbol{\nabla}(\boldsymbol{A}\cdot\boldsymbol{B}) = \boldsymbol{\nabla}\big(A_xB_x + A_yB_y + A_zB_z\big) = \boldsymbol{\nabla}(3xy - 4xy) =$$

$$= \frac{\partial(-xy)}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial(-xy)}{\partial y}\hat{\boldsymbol{y}} = \boxed{-y\hat{\boldsymbol{x}} - x\hat{\boldsymbol{y}}}$$

RHS:

$$\underbrace{\boldsymbol{A} \times (\boldsymbol{\nabla} \times \boldsymbol{B})}_{(1)} + \underbrace{\boldsymbol{B} \times (\boldsymbol{\nabla} \times \boldsymbol{A})}_{(2)} + \underbrace{(\boldsymbol{A} \cdot \boldsymbol{\nabla})\boldsymbol{B}}_{(3)} + \underbrace{(\boldsymbol{B} \cdot \boldsymbol{\nabla})\boldsymbol{A}}_{(4)}$$

$$(1)\ \boldsymbol{A} \times \left[\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right)\hat{\boldsymbol{z}}\right] =$$
$$= (x\hat{\boldsymbol{x}} + 2y\hat{\boldsymbol{y}} + 3z\hat{\boldsymbol{z}}) \times (-5\hat{\boldsymbol{z}}) = -10y\hat{\boldsymbol{x}} + 5x\hat{\boldsymbol{y}}$$

$$(2)\ \boldsymbol{B} \times \left[\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{\boldsymbol{z}}\right] = 0$$

$$(3)\ (\boldsymbol{A} \cdot \boldsymbol{\nabla})\boldsymbol{B} =$$
$$= \left(A_x\frac{\partial B_x}{\partial x} + A_y\frac{\partial B_x}{\partial y} + A_z\frac{\partial B_x}{\partial z}\right)\hat{\boldsymbol{x}}$$
$$+ \left(A_x\frac{\partial B_y}{\partial x} + A_y\frac{\partial B_y}{\partial y} + A_z\frac{\partial B_y}{\partial z}\right)\hat{\boldsymbol{y}}$$
$$+ \left(A_x\frac{\partial B_z}{\partial x} + A_y\frac{\partial B_z}{\partial y} + A_z\frac{\partial B_z}{\partial z}\right)\hat{\boldsymbol{z}} =$$
$$= 6y\hat{\boldsymbol{x}} - 2x\hat{\boldsymbol{y}}$$

$$(4)\ (\boldsymbol{B} \cdot \boldsymbol{\nabla})\boldsymbol{A} =$$
$$= \left(B_x\frac{\partial A_x}{\partial x} + B_y\frac{\partial A_x}{\partial y} + B_z\frac{\partial A_x}{\partial z}\right)\hat{\boldsymbol{x}}$$
$$+ \left(B_x\frac{\partial A_y}{\partial x} + B_y\frac{\partial A_y}{\partial y} + B_z\frac{\partial A_y}{\partial z}\right)\hat{\boldsymbol{y}}$$
$$+ \left(B_x\frac{\partial A_z}{\partial x} + B_y\frac{\partial A_z}{\partial y} + B_z\frac{\partial A_z}{\partial z}\right)\hat{\boldsymbol{z}} =$$
$$= 3y\hat{\boldsymbol{x}} - 4x\hat{\boldsymbol{y}}$$

$$(1) + (2) + (3) + (4) = (-10y + 6y + 3y)\hat{\boldsymbol{x}} + (5x - 2x - 4x)\hat{\boldsymbol{y}} = \boxed{-y\hat{\boldsymbol{x}} - x\hat{\boldsymbol{y}}}$$

(c) LHS:

$$\boldsymbol{\nabla} \times (\boldsymbol{A} \times \boldsymbol{B}) = \boldsymbol{\nabla} \times \left[(A_yB_z - A_zB_y)\hat{\boldsymbol{x}} + (A_zB_x - A_xB_z)\hat{\boldsymbol{y}} + (A_xB_y - A_yB_x)\hat{\boldsymbol{z}}\right] =$$
$$= \boldsymbol{\nabla} \times \left[6xz\hat{\boldsymbol{x}} + 9yz\hat{\boldsymbol{y}} - (2x^2 + 6y^2)\hat{\boldsymbol{z}}\right] = \boldsymbol{\nabla} \times \boldsymbol{C} =$$
$$= \left(\frac{\partial C_z}{\partial y} - \frac{\partial C_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial C_x}{\partial z} - \frac{\partial C_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial C_y}{\partial x} - \frac{\partial C_x}{\partial y}\right)\hat{\boldsymbol{z}} =$$
$$= (-12y - 9y)\hat{\boldsymbol{x}} + (6x + 4x)\hat{\boldsymbol{y}} = \boxed{-21y\hat{\boldsymbol{x}} + 10x\hat{\boldsymbol{y}}}$$

RHS: From (b) it was calculated that $(\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B} = 6y\hat{\boldsymbol{x}} - 2x\hat{\boldsymbol{y}}$ and $(\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A} = 3y\hat{\boldsymbol{x}} - 4x\hat{\boldsymbol{y}}$. The remaining parts of (vi) RHS is

$$\boldsymbol{A}\underbrace{(\boldsymbol{\nabla}\cdot\boldsymbol{B})}_{=0} - \boldsymbol{B}\underbrace{(\boldsymbol{\nabla}\cdot\boldsymbol{A})}_{=6} = -6\boldsymbol{B} = -18y\hat{\boldsymbol{x}} + 12x\hat{\boldsymbol{y}}$$

Summing it all up according to (vi), the result is

$$(3y - 6y - 18y)\hat{\boldsymbol{x}} + (-4x + 2x + 12x)\hat{\boldsymbol{y}} = \boxed{-21y\hat{\boldsymbol{x}} + 10x\hat{\boldsymbol{y}}}$$

□

Problem 1.26

Calculate the Laplacian of the following functions:

(a) $T_a = x^2 + 2xy + 3z + 4$

(b) $T_b = \sin x \sin y \sin z$

(c) $T_c = e^{-5x}\sin 4y \cos 3z$

(d) $\boldsymbol{v} = x^2\hat{\boldsymbol{x}} + 3xz^2\hat{\boldsymbol{y}} - 2xz\hat{\boldsymbol{z}}$

Solution:

The Laplacian of a function T is defined as

$$\boldsymbol{\nabla}^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}$$

and the Laplacian of a vector $\boldsymbol{v}$ is defined as

$$\boldsymbol{\nabla}^2\boldsymbol{v} = (\boldsymbol{\nabla}^2 v_x)\hat{\boldsymbol{x}} + (\boldsymbol{\nabla}^2 v_y)\hat{\boldsymbol{y}} + (\boldsymbol{\nabla}^2 v_z)\hat{\boldsymbol{z}}$$

Using this, the result is

(a) $\boldsymbol{\nabla}^2 T_a = 2$

(b) $\boldsymbol{\nabla}^2 T_b = -3\sin x \sin y \sin z$

(c) $\boldsymbol{\nabla}^2 T_c = e^{-5x}\sin 4y \cos 3z(25 - 16 - 9) = 0$

(d) $\boldsymbol{\nabla}^2\boldsymbol{v} = 2\hat{\boldsymbol{x}} + 6x\hat{\boldsymbol{y}}$

□

Problem 1.27

Prove that the divergence of a curl is always zero.

Solution:

For a vector function $\boldsymbol{v}$, show

$$\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{v}) = 0$$

Using the definition of the divergence and curl:

$$\boldsymbol{\nabla}\cdot\boldsymbol{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

$$\boldsymbol{\nabla}\times\boldsymbol{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\boldsymbol{z}}$$

the result is

$$\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{v}) = \frac{\partial}{\partial x}\left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right) =$$

$$= \frac{\partial^2 v_z}{\partial x\partial y} - \frac{\partial^2 v_y}{\partial x\partial z} + \frac{\partial^2 v_x}{\partial y\partial z} - \frac{\partial^2 v_z}{\partial y\partial x} + \frac{\partial^2 v_y}{\partial z\partial x} - \frac{\partial^2 v_x}{\partial z\partial y} =$$

$$= \underbrace{\frac{\partial^2 v_z}{\partial x\partial y} - \frac{\partial^2 v_z}{\partial y\partial x}}_{=0} + \underbrace{\frac{\partial^2 v_x}{\partial y\partial z} - \frac{\partial^2 v_x}{\partial z\partial y}}_{=0} + \underbrace{\frac{\partial^2 v_y}{\partial z\partial x} - \frac{\partial^2 v_y}{\partial x\partial z}}_{=0}$$

since the symmetry of second derivatives ensures that

$$\frac{\partial^2}{\partial x\partial y} = \frac{\partial^2}{\partial y\partial x}$$

□

Problem 1.28

Prove that the curl of a gradient is always zero.

Solution:

For a function T, show

$$\boldsymbol{\nabla}\times(\boldsymbol{\nabla}T) = 0$$

Using the definitions of the gradient and curl

$$\boldsymbol{\nabla} T = \frac{\partial T}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial T}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial T}{\partial z}\hat{\boldsymbol{z}}$$

$$\boldsymbol{\nabla} \times \boldsymbol{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\boldsymbol{z}}$$

results in

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} T) = \underbrace{\left(\frac{\partial^2 T}{\partial y \partial z} - \frac{\partial^2 T}{\partial z \partial y}\right)}_{=0}\hat{\boldsymbol{x}} + \underbrace{\left(\frac{\partial^2 T}{\partial z \partial x} - \frac{\partial^2 T}{\partial x \partial z}\right)}_{=0}\hat{\boldsymbol{y}} + \underbrace{\left(\frac{\partial^2 T}{\partial x \partial y} - \frac{\partial^2 T}{\partial y \partial x}\right)}_{=0}\hat{\boldsymbol{z}}$$

since the symmetry of second derivatives ensures that

$$\frac{\partial^2}{\partial x \partial y} = \frac{\partial^2}{\partial y \partial x}$$

□

Problem 1.29

Calculate the line integral of the function $\boldsymbol{v} = x^2\hat{\boldsymbol{x}} + 2yz\hat{\boldsymbol{y}} + y^2\hat{\boldsymbol{z}}$ from the origin to the point $(1, 1, 1)$ by three different routes:

(a) $(0, 0, 0) \to (1, 0, 0) \to (1, 1, 0) \to (1, 1, 1)$

(b) $(0, 0, 0) \to (0, 0, 1) \to (0, 1, 1) \to (1, 1, 1)$

(c) The direct straight line

(d) What is the line integral around the closed loop that goes out along path (a) and back along path (b)?

Solution:

The line integral has the from

$$\int_a^b \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{l}$$

where $\boldsymbol{dl} = dx\hat{\boldsymbol{x}} + dy\hat{\boldsymbol{y}} + dz\hat{\boldsymbol{z}}$

(a)

$(0, 0, 0) \to (1, 0, 0)$ has $\boldsymbol{dl} = dx\hat{\boldsymbol{x}}$ and x: $0 \to 1, y = 0, z = 0$.

$(1, 0, 0) \to (1, 1, 0)$ has $\boldsymbol{dl} = dy\hat{\boldsymbol{y}}$ and y: $0 \to 1, x = 1, z = 0$.

$(1,1,0) \to (1,1,1)$ has $d\boldsymbol{l} = dz\hat{\boldsymbol{z}}$ and z: $0 \to 1, x = 1, y = 1$.

$$\int_0^1 x^2\,\mathrm{d}x + \int_0^1 2yz\,\mathrm{d}y + \int_0^1 y^2\,\mathrm{d}z = \int_0^1 x^2\,\mathrm{d}x + \int_0^1 1\,\mathrm{d}z = \frac{1}{3} + 1 = \boxed{\frac{4}{3}}$$

(b)

$(0,0,0) \to (0,0,1)$ has $\mathrm{d}\boldsymbol{l} = \mathrm{d}z\hat{\boldsymbol{z}}$ and z: $0 \to 1, x = 0, y = 0$.

$(0,0,1) \to (0,1,1)$ has $\mathrm{d}\boldsymbol{l} = \mathrm{d}y\hat{\boldsymbol{y}}$ and y: $0 \to 1, x = 0, z = 1$.

$(0,1,1) \to (1,1,1)$ has $\mathrm{d}\boldsymbol{l} = \mathrm{d}x\hat{\boldsymbol{x}}$ and x: $0 \to 1, y = 1, z = 1$.

$$\int_0^1 y^2\,\mathrm{d}z + \int_0^1 2yz\,\mathrm{d}y + \int_0^1 x^2\,\mathrm{d}x = \int_0^1 2y\,\mathrm{d}y + \int_0^1 x^2\,\mathrm{d}x = 1 + \frac{1}{3} = \boxed{\frac{4}{3}}$$

(c) The direct straight line is obtained by setting $x = y = z = t$ and letting t: $0 \to 1$. Also, $\mathrm{d}\boldsymbol{l} = \mathrm{d}t\hat{\boldsymbol{t}}$, giving $\boldsymbol{v} \cdot \mathrm{d}\boldsymbol{l} = 4t^2\,\mathrm{d}t$.

$$\int_0^1 4t^2\,\mathrm{d}t = \boxed{\frac{4}{3}}$$

(d) Since the result of the line integral of $\boldsymbol{v}$ is independent of the path, going from the origin to the point $(1,1,1)$ via the path in (a), and then from the point $(1,1,1)$ to the origin via the path in (b) gives

$$\underbrace{\int_0^1 x^2\,\mathrm{d}x + \int_0^1 1\,\mathrm{d}z}_{(a)} - \left[\underbrace{\int_1^0 2y\,\mathrm{d}y + \int_1^0 x^2\,\mathrm{d}x}_{(b)\text{ in reverse}}\right] = \boxed{\frac{4}{3} - \frac{4}{3} = 0}$$

□

Problem 1.30

Calculate the surface integral of the function

$$\boldsymbol{v} = 2xz\hat{\boldsymbol{x}} + (x+2)\hat{\boldsymbol{y}} + y(z^2 - 3)\hat{\boldsymbol{z}}$$

over the bottom of the box. For consistency, let "upward" be the positive direction. Does the surface integral depend only on the boundary line for this function? What is the total flux over the closed surface of the box (including the bottom)?

Solution:

For the bottom of the box, x, y: $0 \to 2$, $z = 0$ and $\mathrm{d}\boldsymbol{a} = \mathrm{d}x\,\mathrm{d}y\hat{\boldsymbol{z}}$

$$\int_0^2 \int_0^2 -3y\,\mathrm{d}x\,\mathrm{d}y = \int_0^2 \left(\int_0^2 -3y\,\mathrm{d}y \right) \mathrm{d}x = \int_0^2 -6\,\mathrm{d}x = \boxed{-12}$$

The total outward flux from all faces exluding the bottom was 20 (result from Ex. 1.7). The resulting total outward flux is then $20 - (-12) = 32$ (since the result from the bottom surface was calculated in the "wrong" direction). The surface integral does not depend only on the boundary line, since the bottom surface shares the boundary line with the combined 5 other surfaces, and the calculations gave different values (20 for the 5 surfaces, -12 for the bottom surface).

□

Problem 1.31

Problem 1.32

Problem 1.33

Problem 1.34

Problem 1.35

Problem 1.36

Problem 1.37

Problem 1.38

Problem 1.39

Problem 1.40

Problem 1.41

Problem 1.42

Problem 1.43

Problem 1.44

Problem 1.45

Problem 1.46

Problem 1.47

Problem 1.48

Problem 1.49

Problem 1.50

Problem 1.51

Problem 1.52

Problem 1.53

Problem 1.54

Problem 1.55

Problem 1.56

Problem 1.57

Problem 1.58

Problem 1.59

Problem 1.60

Problem 1.61

Problem 1.62

Problem 1.63

Problem 1.64

Problem 1.65